

SECTION B

9 **(long)**

(a) Find the general solution to

$$\frac{d^2x}{dt^2} + (2 + \alpha)\frac{dx}{dt} + 2\alpha x = f(t), \quad (1)$$

where $f(t) = 0$ and $\alpha = 3$. [10]

(b) Find the general solution to equation (1) for any α , subject to $f(t) = 0$ and $x(0) = 1$. [10]

(c) Find the general solution to equation (1) for $\alpha = 2$, subject to $f(t) = 1 + t$ and $x(0) = 0$. [10]

10 **(long)** Consider the two coupled difference equations

$$\begin{aligned} x_{n+1} &= \alpha x_n + \frac{1}{2}y_n \\ y_{n+1} &= (1 - \alpha)x_n + \frac{1}{2}y_n, \end{aligned}$$

where $0 \leq \alpha \leq 1$ is a fixed parameter.

(a) Show that the difference equation can be written in the form $\mathbf{z}_{n+1} = A\mathbf{z}_n$, where $\mathbf{z}_n = [x_n \ y_n]^t$. State the matrix A . [6]

(b) If $\mathbf{z}_0 = [1 \ 0]^t$, compute $\mathbf{z}_n$ as $n \rightarrow \infty$. You may use the fact that the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = \alpha - \frac{1}{2}$. [12]

(c) Now assume that $\mathbf{z}_0$ is a unit vector and we observe that $\mathbf{z}_2 = [0 \ 0]^t$. Calculate α . [12]